# SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

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## Mini Project Synopsis

## "AI-Powered Banking Fraud & Compliance Intelligence Platform"

submitted in the partial fulfilment of the requirements for V semester, Bachelor of Engineering

**in**

**COMPUTER SCIENCE & ENGINEERING**

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# Chapter 1

# Introduction

Banks and financial institutions process hundreds of millions of transactions every single day. Hidden inside this enormous volume are fraudsters, money launderers, identity thieves, and organised criminal networks, all attempting to move illegitimate funds or steal from genuine customers. The scale of the problem is significant: banking fraud is estimated to cost more than $45 billion annually worldwide, cross-border payment volumes are projected to exceed $200 trillion by 2025, and regulators impose roughly $5 billion in fines every year on banks that fail to meet Anti-Money Laundering (AML) obligations. In the United States alone, consumers reported losing $12.5 billion to fraud in 2024.

Despite this scale, most banks continue to rely on *rule-based* fraud detection systems, in which engineers manually author static rules such as "if a transaction exceeds a fixed amount *and* the account is newly opened *and* the recipient is overseas, flag it for review." While simple to implement, this approach is fundamentally reactive: it only catches fraud after a transaction has already been attempted, it cannot adapt to new criminal tactics without manual rule updates, and it treats each transaction in isolation rather than as part of a broader behavioural or relational pattern. In practice, over 95% of the alerts generated by such systems turn out to be false positives, meaning analysts spend the majority of their time investigating entirely legitimate activity while genuine fraud can slip through amid the noise.

A second, equally important limitation of existing systems is data fragmentation. A bank typically holds transaction histories, Know-Your-Customer (KYC) records, login and device behaviour, geolocation data, and watchlist information, but these data sources are usually managed by separate teams using separate systems that rarely communicate with one another. Fraud investigators look at transactions, compliance teams look at KYC documentation, and security teams look at login and device data, but no single view brings all of this together. This is precisely where sophisticated fraud rings and money-laundering networks hide: an individual account may look completely normal in isolation, while sharing a phone number, device fingerprint, or IP address with several

other accounts reveals a coordinated network of criminal activity.

This project proposes an AI-powered platform that addresses both limitations directly. Instead of evaluating a transaction in isolation against static thresholds, the platform unifies transaction, KYC, and behavioural data into a single knowledge graph, allowing hidden relationships between seemingly unrelated accounts to be surfaced. On top of this unified view, an AI-based risk-scoring model evaluates each transaction in real time using both its own features and the surrounding network context, producing an explainable risk score rather than a single-rule pass/fail decision. Finally, once a transaction or account is flagged, a Retrieval-Augmented Generation (RAG) based compliance assistant automatically drafts a Suspicious Activity Report (SAR) grounded in the actual regulatory text relevant to the case, reducing a task that otherwise takes an analyst two to four hours down to a matter of minutes.

The central shift this project makes is moving away from asking "is this transaction, on its own, suspicious?" towards asking "does this account, in the context of its entire network of relationships and behavioural history, match a known criminal pattern?" This entity-level, network-aware framing, combined with a real-time, explainable, and RAG-grounded compliance pipeline, forms the basis of the design discussed throughout the remainder of this report.

# Chapter 2

# Literature Survey

A review of recent research (2024–2026) was carried out across three themes directly relevant to this project: graph-based fraud detection, machine-learning-driven Anti-Money-Laundering (AML) systems, and explainable or LLM/RAG-based compliance tooling. Fourteen peer-reviewed and pre-print works were examined in total. Full bibliographic details, including author names, are provided in the References chapter; this section discusses each work by its title and contribution.

## 2.1 Graph Neural Networks for Fraud Detection

A substantial body of recent work applies Graph Neural Networks (GNNs) to financial fraud detection, motivated by the observation that fraud rarely occurs in isolation and is better modelled as a pattern across a network of accounts, devices, and transactions rather than as an isolated data point. *"Graph Neural Networks for Financial Fraud Detection: A Review"* [1] surveys the landscape of GNN architectures applied across fraud domains, establishing a taxonomy of node-level, edge-level, and graph-level fraud-detection tasks. Building on this foundation, *"CaT-GNN: Causal Temporal Graph Neural Networks for Credit Card Fraud Detection"* [2] introduces causal and temporal reasoning into the graph representation, allowing the model to distinguish genuine causal transaction patterns from coincidental structural similarity, and reports improved performance over prior baselines on both private and public credit-card datasets. Similarly, *"Detecting Credit Card Fraud via Heterogeneous Graph Neural Networks with Graph Attention"* [3] models transaction networks as heterogeneous graphs containing multiple node types – users, merchants, and transactions – and applies graph attention to weigh the influence of different neighbour types rather than treating every connection equally. *"Harnessing Graph Neural Networks for Enhanced Fraud Detection"* [4] benchmarks GNN-based approaches directly against traditional machine-learning baselines and reports consistent gains in detecting coordinated, multi-account fraud, while *"TMR-GGNN: Time-Aware Multi-Relational Guided Graph Neural Network"* [5] further extends this line of work with a time-aware, multi-

relational GNN that explicitly encodes the temporal ordering of transactions alongside multiple relationship types (e.g. shared device, shared recipient) between accounts.

## 2.2 AML and Money-Laundering Detection

A second cluster of literature addresses Anti-Money-Laundering detection specifically, often with an emphasis on reducing the false-positive burden that plagues real-world compliance teams. *"Enhancing Anti-Money Laundering Protocols: Employing Machine Learning to Minimise False Positives and Improve Operational Cost Efficiency"* [6], presented at the 2024 International Conference on Advances in Artificial Intelligence, proposes a machine-learning pipeline explicitly aimed at reducing false-positive AML alerts, directly addressing the problem statement of this project. A systematic literature review, *"Machine Learning Models for Money Laundering Detection in Financial Institutions"* [7], presented at LACCEI 2024, consolidates the range of machine-learning techniques applied to money-laundering detection across the recent literature, while *"Comparative Analysis of Machine Learning Algorithms for Money Laundering Detection"* [8] provides a comparative benchmark of several classifiers (including gradient-boosted trees and neural approaches) for suspicious-transaction detection. *"Leveraging Artificial Intelligence for Enhanced Anti-Money Laundering in Financial Institutions"* [9] reviews the practical adoption patterns of AI within bank AML programmes, highlighting the gap between what is technically demonstrated in research and what is operationally deployed. Finally, *"DPxFin: Adaptive Differential Privacy for Anti-Money Laundering Detection via Reputation-Weighted Federated Learning"* [10] addresses a complementary concern – data privacy – by proposing a federated-learning approach to AML detection that allows multiple institutions to collaboratively train a detection model without sharing raw customer data directly.

## 2.3 Explainability and LLM/RAG-Based Compliance

The third theme concerns explainability and the emerging use of large language models (LLMs) in regulatory and compliance workflows. *"Secure and Transparent Banking: Explainable AI-Driven Federated Learning for Fraud Detection"* [11] combines federated learning with SHAP- and LIME-based explainability techniques to produce fraud-detection decisions that remain both privacy-preserving and interpretable to a human reviewer. *"Financial Fraud Detection Using Explainable AI and Stacking Ensemble Meth-*

*ods"* [12] applies a stacking ensemble alongside SHAP explanations, demonstrating that ensemble accuracy gains need not come at the cost of interpretability. *"A User-Centered Explainable AI Approach for Financial Fraud Detection"* [13] takes a user-centred approach, designing explanations specifically around the practical needs of a fraud analyst rather than around what is mathematically convenient to compute. Most relevant to the compliance-automation component of this project, *"FinGuard: Detecting Financial Regulatory Non-Compliance in LLM Interactions"* [14] proposes a system that uses LLMs to detect regulatory non-compliance within financial workflows, illustrating both the promise and the current limitations of applying generative AI directly to regulatory text.

## 2.4 Limitations of the Existing Works

While the reviewed literature demonstrates strong individual results, several consistent gaps emerge when the works are considered together, and these gaps directly motivate the design of this project.

- **Detection models are studied in isolation.** The GNN-based fraud-detection papers [1–5] each optimise a single detection model and evaluate it against a fixed benchmark dataset. None of them integrate the detection model into a broader investigation or compliance workflow – a trained model that outputs a risk score is only one part of what a real fraud or compliance team actually needs.

- **AML systems remain largely siloed per institution.** The AML-focused works [6–10] reduce false positives and improve detection accuracy, but generally operate on a single institution's transaction data without a shared, cross-source knowledge graph that links transactions to KYC records, device fingerprints, and behavioural history in one structure.

- **Explainability is applied to one model at a time.** The XAI-focused papers [11–13] add SHAP- or LIME-based interpretability to a single classifier, but none extend this to a multi-stage decision pipeline, where an analyst may need an explanation that spans several models and a knowledge-graph traversal, not just one.

- **LLM/RAG-based compliance tooling is not integrated with real-time detection.** FinGuard [14] demonstrates that LLMs can be applied to regulatory-

compliance checking, but this and similar tools operate as a separate layer, disconnected from the underlying fraud or AML detection models that would actually trigger a compliance review.

- **Validation is largely restricted to public or synthetic benchmark datasets** (for example, the widely-used Kaggle credit-card fraud dataset), rather than at the scale and complexity of a real bank's multi-source data. This is an acknowledged limitation of the present project as well, and is discussed further in Chapter 6.

Taken together, no reviewed work combines a unified, cross-source knowledge graph, a real-time and explainable risk-scoring model, and a RAG-grounded compliance-automation pipeline within a single deployable system. This combination is precisely the gap that the proposed platform, described in Chapter 5, is designed to address.

# Chapter 3

# Problem Statement

Most banks today continue to rely on rule-based fraud detection systems, in which engineers manually author static rules such as "flag any transaction above a fixed amount, from an account less than 30 days old, to a recipient in a foreign country." While such rules are simple to implement and audit, they suffer from three fundamental flaws that this project directly addresses.

## 3.1 Overwhelming False Positives

Rule-based systems are notoriously imprecise: over 95% of all alerts they generate turn out to be false positives, flagging entirely legitimate, innocent transactions. Each such alert still requires a human analyst to investigate, typically taking 30 to 70 minutes per case. The consequences are severe: analysts are buried in low-value investigative work, genuine customers experience frozen accounts and declined transactions for entirely ordinary behaviour, and – most critically – real fraud is more likely to slip through because investigative teams are overwhelmed by noise. Industry estimates place the cost of investigating false positives at approximately $3.5 billion per year across the banking sector.

## 3.2 Purely Reactive Detection

A rule-based system can only ever react *after* a suspicious transaction has already been attempted; it has no mechanism for recognising that an account is gradually being prepared for fraudulent use over days or weeks based on accumulating behavioural signals. By the time a rule fires, the transaction – and often the underlying funds – may already be in motion. There is no equivalent of a predictive, early-warning capability in a purely rule-based architecture.

## 3.3 Completely Siloed Data

A bank holds an enormous amount of relevant data about each customer: transaction history, KYC documentation, login and device behaviour, geolocation, and customer-service

interactions. In practice, these data sources sit in separate systems, owned by separate teams, that rarely communicate with one another. The fraud team examines transactions; the compliance team examines KYC records; the security team examines login and device data – but no single view brings all of this together. This fragmentation is precisely where sophisticated criminal networks operate undetected: an individual account can look entirely unremarkable in isolation, while sharing a phone number, device fingerprint, or IP address with several other seemingly unrelated accounts reveals a coordinated fraud ring or money-laundering network that no single, siloed system can see.

In summary, the problem this project addresses is: *existing fraud and AML systems are reactive, imprecise, and structurally blind to cross-account relationships, resulting in high operational cost, poor customer experience, and continued exposure to sophisticated, network-based financial crime.*

# Chapter 4

# Objectives

Based on the problem statement and the scope achievable within the mini-project timeline, the following three objectives have been identified as the core, achievable deliverables of this project:

1. **Build a unified knowledge graph** that links transaction, KYC, and behavioural data for a chosen set of accounts, so that relationships between accounts (shared devices, shared phone numbers, shared recipients) that are invisible in a purely tabular view become directly queryable.

2. **Implement an AI-based transaction risk-scoring model** (gradient-boosted-tree and/or Graph-Neural-Network driven) that flags suspicious transactions in real time, using both the transaction's own features and context derived from the knowledge graph, together with an explanation for each score produced.

3. **Automate generation of a compliant Suspicious Activity Report (SAR) draft** from flagged cases using a Retrieval-Augmented Generation (RAG) pipeline grounded in real regulatory text, reducing the manual drafting effort currently required of a compliance analyst.

These three objectives were deliberately scoped to be independently achievable and independently demonstrable, so that the project delivers a working, evaluable result on each even if time constraints limit further extension. Additional capabilities envisioned for a full production platform – such as a dedicated AML network-analysis agent, a continuously updated customer risk profiler, and an interactive fraud-investigation graph UI – are discussed as part of the proposed design in Chapter 5, but are treated as stretch goals beyond the three core objectives above.

# Chapter 5

# Proposed Design (If any)

The proposed platform is organised into four layers, each feeding into the next, as shown in Figure 5.1.

1. **Layer 1 – Data Ingestion:** raw transaction streams, KYC profiles, behavioural logs (login, geolocation, device), and external watchlists are ingested in real time.

2. **Layer 2 – Intelligence Engine:** ingested data is unified into a knowledge graph capturing entities (accounts, devices, phone numbers) and their relationships, alongside a Retrieval-Augmented Generation (RAG) and LLM core used for regulatory-document retrieval and narrative generation.

3. **Layer 3 – AI Agent Layer:** a set of specialised components operate on top of the shared knowledge graph and intelligence engine – a Fraud Detection Engine (GNN and/or anomaly-detection driven), an AML Intelligence Agent for cross-account network analysis, a Transaction Risk Scoring component, a Compliance Assistant for automated SAR drafting, a Customer Risk Profiler, and a Fraud Investigation Graph for analyst-facing visualisation. Of these, the Fraud Detection/Risk Scoring component and the Compliance Assistant correspond directly to Objectives 2 and 3; the remaining components are discussed as extensions in Section 5.3.

4. **Layer 4 – Actions and Outputs:** the analyst dashboard, automatic transaction blocking/freezing, regulatory filing, and case-management workflow that surface the platform's outputs to a human analyst.

A feedback loop connects analyst decisions (confirmed fraud, cleared false positives) back into Layer 1, so that the underlying models are continuously retrained on real outcomes rather than remaining static after initial training.

## 5.1 End-to-End Flow

Figure 5.2 traces a representative scenario – a money-mule account being used to move funds overseas – through all four layers of the platform, from the initial transaction

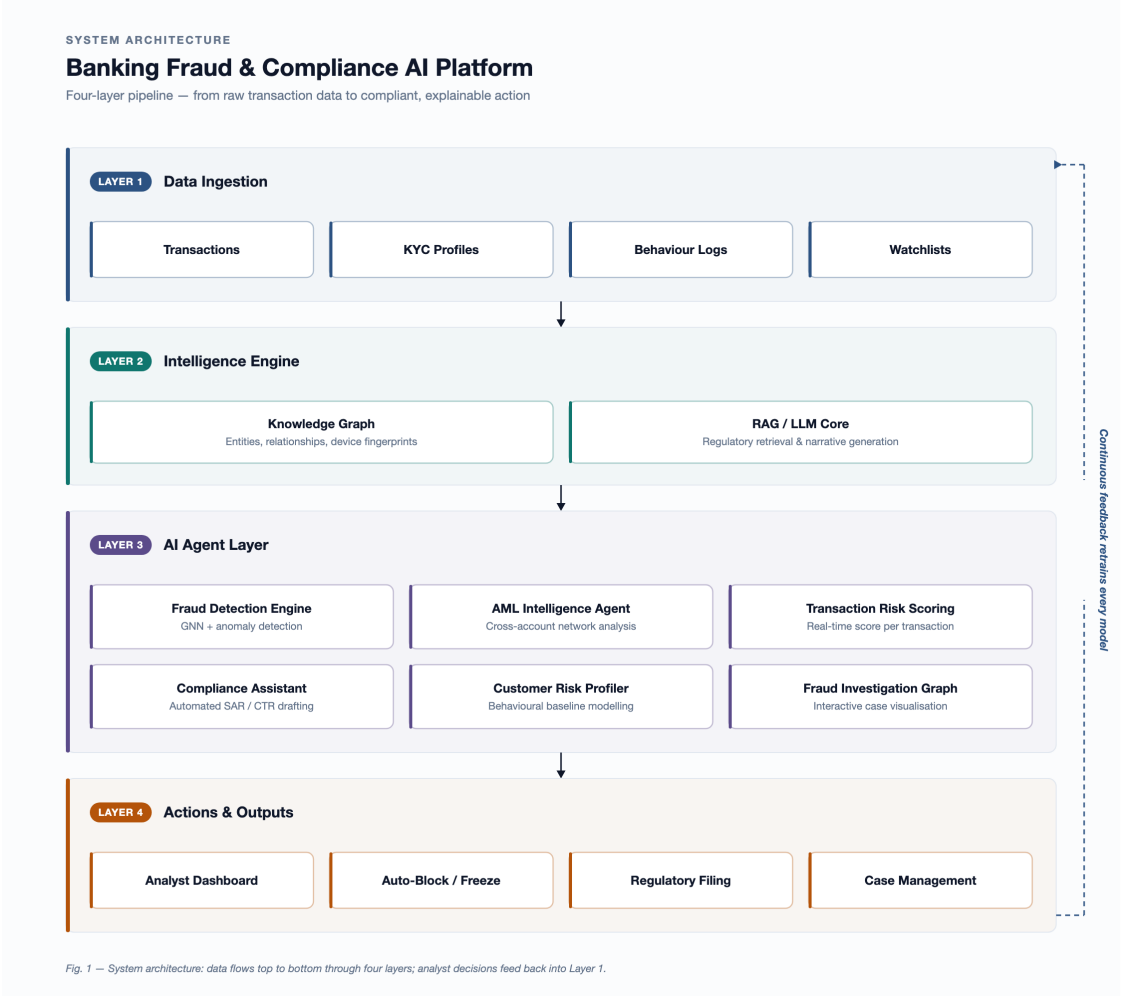



Figure 5.1: Four-layer system architecture of the proposed platform

through to a closed, compliant case.

1. **Transaction initiated:** an overseas wire transfer is streamed in real time into the intelligence engine.

2. **Risk score computed:** the Transaction Risk Scoring component evaluates the transaction against the account's behavioural baseline (device novelty, time-of-day deviation, amount deviation), producing a preliminary risk score.

3. **Knowledge graph queried:** the knowledge graph is queried for the account's connections, revealing that its registered phone number and device fingerprint are shared with several other recently opened accounts – raising the risk score substantially.

4. **AML network analysis:** the AML Intelligence Agent maps the full network of connected accounts and recognises a known layering/structuring pattern, where funds are moved through several accounts in amounts just below regulatory reporting thresholds.

5. **Transaction auto-frozen:** the transaction is automatically blocked before completion, and the case is pushed to the analyst dashboard with a pre-built investigation summary.

6. **SAR auto-generated:** the Compliance Assistant retrieves the relevant regulatory text via RAG and drafts a complete SAR narrative, citing the specific transactions and regulatory clauses that justify the filing.

7. **Analyst review and feedback:** the analyst reviews the visualised network and the generated SAR, confirms the case, and this outcome is logged to retrain the underlying models for future cases.

## 5.2 Scope for the Present Mini-Project

Given the timeline available for this mini-project, implementation effort is prioritised on the three components that map directly to the objectives stated in Chapter 4: the knowledge graph (Layer 2), the transaction risk-scoring model (part of Layer 3), and the RAG-based Compliance Assistant (part of Layer 3). The remaining Layer 3 components

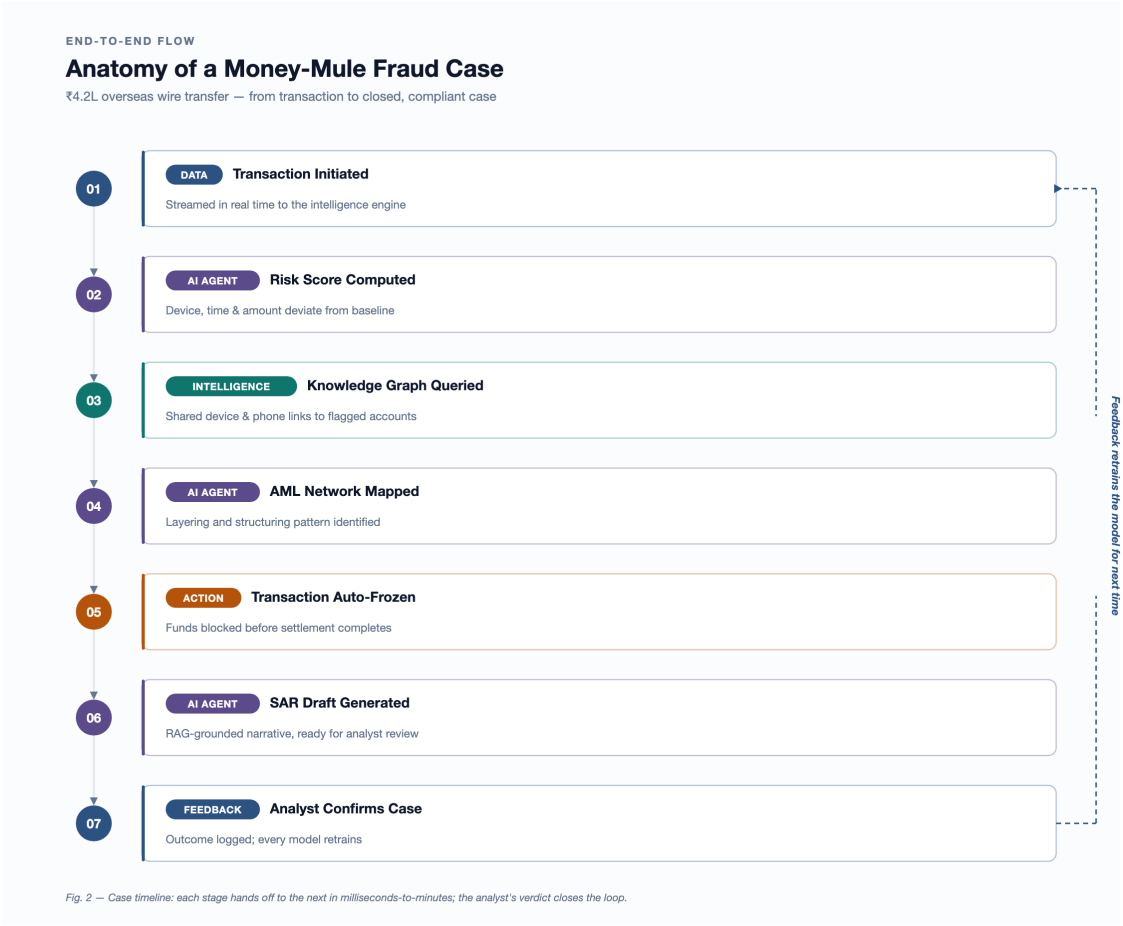



Figure 5.2: End-to-end flow for a representative money-mule fraud case

– the dedicated AML network agent, the continuously updated customer risk profiler, and the interactive fraud-investigation graph UI – along with full production-grade deployment (Layer 4 automation, containerised infrastructure), are treated as extensions beyond the current scope and are discussed as future work in Chapter 6.

# Chapter 6

# Expected Outcome

At the completion of this mini-project, the following concrete, evaluable outcomes are expected:

1. A working **knowledge graph**, populated with transaction, KYC-style, and behavioural data for a representative set of accounts, capable of answering investigative queries such as "which other accounts share this device or phone number" and "what is the shortest connecting path between two flagged accounts," including at least one deliberately constructed fraud-ring scenario that is demonstrably detectable through the graph.

2. A trained **transaction risk-scoring model**, evaluated using precision, recall, F1-score, and area-under-precision-recall-curve on a time-respecting train/test split (rather than accuracy, which is not meaningful for the highly imbalanced class distribution typical of fraud data), together with SHAP-based explanations for individual predictions so that any flagged transaction can be justified in human-readable terms.

3. A working **Compliance Assistant** that, given a flagged case, retrieves relevant excerpts from a small corpus of real regulatory documents (such as PMLA 2002 provisions and RBI KYC guidance) and generates a structured, cited Suspicious Activity Report draft, with citations manually verified against the source documents to confirm the system is not fabricating regulatory references.

4. A **demonstration interface** tying the above three components into a single coherent flow – a transaction receives a risk score with explanation, its network context is visualised from the knowledge graph, and a SAR can be generated for the flagged case – exercised through a small number of prepared, realistic demo scenarios.

## 6.1 Expected Impact

If the three core objectives are met, the resulting prototype is expected to demonstrate, at a small but representative scale, the central claim motivating this project: that a unified, network-aware, explainable approach can substantially reduce false-positive investigation load compared to static rule-based detection, while surfacing coordinated fraud patterns (shared devices, shared identifiers across accounts) that a purely transaction-level system would miss entirely, and while reducing the manual effort required to produce a compliant SAR from hours to minutes.

## 6.2 Known Limitations and Future Work

As acknowledged in Chapter 2, the present project is validated on public and synthetically augmented data rather than real bank-scale transaction volumes, and the regulatory document corpus used by the Compliance Assistant is necessarily a small representative subset rather than a comprehensive regulatory library. The full six-component agent architecture envisioned in Chapter 5 – including a dedicated AML network-analysis agent, a continuously updated customer risk profiler, and a fully interactive fraud-investigation graph interface – is likewise treated as future work beyond the three core objectives delivered here. These limitations are stated explicitly so that the results of this project are interpreted at the scale and scope they were actually evaluated at, with a clear path identified for extending the prototype toward a production-grade system.

# Bibliography


[1] "Graph Neural Networks for Financial Fraud Detection: A Review," arXiv:2411.05815, 2024. Available: https://arxiv.org/abs/2411.05815

[2] "CaT-GNN: Causal Temporal Graph Neural Networks for Credit Card Fraud Detection," arXiv:2402.14708, 2024. Available: https://arxiv.org/abs/2402.14708

[3] "Detecting Credit Card Fraud via Heterogeneous Graph Neural Networks with Graph Attention," arXiv:2504.08183, 2025. Available: https://arxiv.org/abs/2504.08183

[4] "Harnessing Graph Neural Networks for Enhanced Fraud Detection," arXiv:2503.22681, 2025. Available: https://arxiv.org/abs/2503.22681

[5] "TMR-GGNN: Time-Aware Multi-Relational Guided Graph Neural Network for Credit Card Fraud Detection," arXiv:2606.18444, 2026. Available: https://arxiv.org/abs/2606.18444

[6] "Enhancing Anti-Money Laundering Protocols: Employing Machine Learning to Minimise False Positives and Improve Operational Cost Efficiency," Proceedings of the 2024 8th International Conference on Advances in Artificial Intelligence (ICAAI), ACM, 2024. Available: https://dl.acm.org/doi/full/10.1145/3704137.3704156

[7] "Machine Learning Models for Money Laundering Detection in Financial Institutions: A Systematic Literature Review," Proceedings of the 22nd LACCEI International Multi-Conference for Engineering, Education, and Technology, 2024. Available: https://laccei.org/LACCEI2024-CostaRica/papers/Contribution_1682_final_a.pdf

[8] "Comparative Analysis of Machine Learning Algorithms for Money Laundering Detection," Discover Artificial Intelligence (Springer), vol. 5, 2025. Available: https://link.springer.com/article/10.1007/s44163-025-00397-4

[9] "Leveraging Artificial Intelligence for Enhanced Anti-Money Laundering in Financial Institutions," Discover Computing (Springer Nature), 2026. Available: https://link.springer.com/article/10.1007/s10791-026-10142-9

[10] "DPxFin: Adaptive Differential Privacy for Anti-Money Laundering Detection via Reputation-Weighted Federated Learning," arXiv:2603.19314, 2026. Available: https://arxiv.org/abs/2603.19314

[11] "Secure and Transparent Banking: Explainable AI-Driven Federated Learning Model for Financial Fraud Detection," Journal of Risk and Financial Management, MDPI, vol. 18, no. 4, p. 179, 2025. Available: https://www.mdpi.com/1911-8074/18/4/179

[12] "Financial Fraud Detection Using Explainable AI and Stacking Ensemble Methods," arXiv:2505.10050, 2025. Available: https://arxiv.org/abs/2505.10050

[13] "A User-Centered Explainable AI Approach for Financial Fraud Detection," Research in International Business and Finance, ScienceDirect, 2024. Available: https://www.sciencedirect.com/science/article/abs/pii/S1544612323006815

[14] "FinGuard: Detecting Financial Regulatory Non-Compliance in LLM Interactions," arXiv:2605.29427, 2026. Available: https://arxiv.org/abs/2605.29427